E-commerce CDC Pipeline — Design & Decision Note

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Objective

Incrementally apply customer/order inserts and partial updates, preserve current state, make reruns safe, and summarize the evidence from each batch.

Architecture

Synthetic JSON events → validation → raw_events immutable ledger → customers/orders current-state tables → change_summaries

Key decisions

- SQLite keeps setup reproducible with no external database service.
- Each event has a stable event_id. raw_events uses it as a primary key; replayed IDs are skipped.
- Partial updates merge into existing rows so fields absent from an update remain unchanged.
- Each state mutation and its raw-event insert are committed in one transaction; invalid events are rolled back.
- Orders require an existing customer; unknown-entity updates and negative order amounts are rejected.
- SummaryProvider isolates AI logic. An optional local Ollama adapter is included; the default offline fallback is clearly labelled and reports only observed counts/examples. No secrets are committed.

Validation

Four automated tests cover insert/update state, idempotent replay, update to a missing customer, and negative order amount. Run: **python -m unittest discover -s tests -v**. Run the pipeline twice to demonstrate first-run processing and second-run deduplication.

Limitations / next steps

This is a local batch simulation, not a production CDC connector. Source event ordering is assumed; production use should add source sequence/version handling, a durable queue/connector, monitoring and a production database. Local LLM summaries may be inaccurate and should be reviewed.